Algebra II with Trigonometry Chapter 7.4 & 7.5

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file: 4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf

1. **One-Period Sine** Solve $\sin\theta = \frac{1}{2}$ for $0 \le \theta < 2\pi$.

2. **All Tangent Solutions** Find all solutions to $\tan\theta = -1$.

3. **Zero-Product Idea** Solve $(2\sin\theta - 1)(\cos\theta) = 0$ for $0 \le \theta < 2\pi$.

4. **Quadratic in Sine** Solve $2\sin^2\theta + 5\sin\theta - 12 = 0$ for $0 \le \theta < 2\pi$.

5. **Use a Pythagorean Identity** Solve $\sin^2\theta = 1 - \cos^2\theta$. What does this tell you about θ?

6. **Double Angle First** Solve $\sin(2\theta) = 0$ for $0 \le \theta < 2\pi$.

7. **Solve for the Multiple Angle** Solve $\tan(3\theta) = 1$ for $0 \le \theta < 2\pi$.

8. **Simple Identity Rewrite** Solve $2\tan^2\theta + \sec^2\theta = 4$ for $0 \le \theta < 2\pi$.

9. **Factor Out a Trig Function** Solve $4\cos\theta\sin\theta + 3\cos\theta = 0$ for $0 \le \theta < 2\pi$.

10. **Recognize a Formula** Solve $\sin 2\theta\cos\theta - \cos 2\theta\sin\theta = \frac{\sqrt{3}}{2}$ for $0 \le \theta < 2\pi$.